Lösungen · sin, cos, tan

1. **a, b · c**

2. **$\sin\gamma = u/w$**

3. **a) sin**
b) cos

4. **$\alpha = \sin^{-1}(7{,}3/13{,}2) \approx 33{,}6°$** ..... $\sin\alpha = \frac{7{,}3}{13{,}2}$

5. **$\beta_1 = \cos^{-1}(\frac{262}{396}) \approx 48{,}6°$** ..... $\cos\beta_1 = \frac{262}{396}$

6. **$\varepsilon = \cos^{-1}(\frac{16}{38}) \approx 65°$** ..... $\cos\varepsilon = \frac{16}{38}$

7. **a) $a = c \cdot \sin\alpha$**
b) $c = \frac{a}{\sin\alpha}$

8. **$a = \frac{7{,}3}{\sin 49°} \approx 9{,}7$ m** ..... $\sin 49° = \frac{7{,}3}{a}$

9. **$AC = \frac{34{,}47}{\sin 24°} \approx 84{,}7$ cm** ..... $\sin 24° = 34{,}47 : AC$

10. **a) BF = AF** ..... $BF = 128{,}9 \cdot \sin 45°$
b) BC = BF : cos 58° ≈ 172,0 cm

11. **a) Pythagoras** ..... Kommt ein Winkel vor, dann sin, cos, tan
b) sin, cos, tan
c) sin, cos, tan

12. **a) $BC = \sqrt{34} \approx 5{,}83$** ..... $AB = 3$, $AC = 5$
b) $\beta = \tan^{-1}(\frac{5}{3}) \approx 59{,}0°$ ..... $\tan\beta = \frac{5}{3}$